AstraZeneca

CONFIDENTIAL AND PROPRIETARY

TECHNICAL REPORT
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SIGNATURES

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TABLE OF CONTENTS

SIGNATURES....1
1 INTRODUCTION....3
1.1 Purpose....3
1.2 Assumptions....4
1.3 Abbreviations/Acronyms/Definition....5
1.4 Definitions of Terminologies for Parameter Ranges....6
1.4.1Target....6
1.4.2Normal Operating Range (NOR)....6
2 BILL OF MATERIALS....7
3 IN-PROCESS HOLD TIMES....8
4 LOAD AND PRODUCT FILTRATION....9
5 CATION EXCHANGE CHROMATOGRAPHY....10
6 AEXMM CHROMATOGRAPHY LOAD ADJUSTMENT....12
7 AEXMM CHROMATOGRAPHY....13
8 HIC LOAD ADJUSTMENT....15
9 HIC CHROMATOGRAPHY....16
10 CONCENTRATION AND DIAFILTRATION....18
11 BULK FILTRATION....20
12 BUFFERS AND SOLUTIONS....21
13 REFERENCE CHROMATOGRAMS....29
14 EDITION HISTORY....34

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1 INTRODUCTION

1.1 Purpose

This report describes the purification process and process parameters for the production of AZD1402 Drug Substance. The overall process (upstream/downstream) is designated "Process 1".

Table 1.1-1 General Properties: AZD1402 Drug Substance

Property	Description
Target	IL-4Rα
Protein classification	Anticalin
Isoelectric point (pI) range (experimental)	6.7
Extinction coefficient	1.47 (theoretical)
Expected titer range (g/L)	1.5 (range: 1.0-2.0)
Overall expected yield from harvest through Drug Substance (%)	40-50
Drug Substance concentration (g/L)	55
Drug Substance formulation	4 mM Sodium/Potassium Phosphate, 155 mM Sodium Chloride, pH 7.0
Drug substance presentation	Stored frozen at ≤ - 80°C in 1 L or 4L HDPE bottles

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1.2 Assumptions

- Unless otherwise noted, all conductivity values are compensated to a reference temperature of 25°C using a temperature coefficient (θ) of 2.0% per °C.
- Unless otherwise noted, flow rate ranges are for steady state operation and do not consider pump ramping strategy.
- Unless otherwise noted, all unit operations are conducted at controlled room temperature (18-26°C).
- Unless otherwise noted, all chromatography step phases are operated in down flow mode.
- Unless otherwise noted, all parameters are inputs.

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1.3 Abbreviations/Acronyms/Definition

Table 1.3-1 Abbreviations and Acronyms

Acronym	Definition	Acronym	Definition
AEXMM	Anion exchange mixed mode chromatography	NOR	Normal operating range
CEX	Cation exchange chromatography	PES	Polyethersulfone
CM	Conditioned media	PFB	Pre-formulated bulk
Conc.	Concentration	P/N	Part number
CV	Column volume	psi; psig	pounds per square inch; pounds per square inch gauge
DV	Diavolumes	RP-HPLC	Reverse phase high performance chromatography
DBC	Dynamic binding capacity	SS	Stainless steel
DS	Drug substance	TBD	To be determined
HIC	Hydrophobic interaction chromatography	TMP	Trans-membrane pressure
kDa	kilodalton	UF/DF	Ultrafiltration diafiltration
LG	Linear gradient	WFI	Water for injection
NA	Not applicable		

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1.4 Definitions of Terminologies for Parameter Ranges

1.4.1 Target

The target value is a preset value that a process parameter is aimed to by process controls. The controls include any combination of manual or automated, discrete or continuous, feed-forward or feed-back responses to operate the process near the target value. Where Targets are provided, the process should be controlled to meet the target value as closely as possible, not just controlling to the edge of the Normal Operating Range.

1.4.2 Normal Operating Range (NOR)

Operating parameter ranges, which are established based on current understanding and process knowledge gained through development and process characterization. NORs are specified in manufacturing batch records and/or other GMP documentation (such as automation recipes) to control the process. A value outside a NOR triggers a non-conformance investigation and is documented and evaluated according to the site's Quality System.

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2 BILL OF MATERIALS

Table 2-1 Bill of Materials

Type	Component Name	Minimum Grade	Catalogue no.	Justification/Notes
Chromatography media	GigaCap CM-650M (CEX)	NA	21948 21949 21950	1 L 5 L 50 L
	Capto Adhere ImpRes (AEXMM)	NA	17371503 17371504	1 L 5 L
	Capto Phenyl (high sub) (HIC)	NA	17545103 17545104	1 L 5 L
Buffer and solution components	Acetic acid, glacial	USP, EP		NA
	Ammonium sulfate	USP, EP		NA
	Citric Acid, monohydrate	USP, EP		NA
	Sodium citrate dihydrate	USP, EP		NA
	Ethanol, 200 proof, Absolute (Dehydrated Alcohol)	NA		NA
	L-Arginine	USP, EP		NA
	Magnesium chloride hexahydrate	USP, EP		NA
	1X Phosphate buffer saline, pH 7.4	USP, EP		NA
	Sodium chloride	USP, EP		NA
	10N Sodium hydroxide	NA		Biotech Reagent
	Sodium hydroxide, pellets	EP, NF		NA
	Tris base	USP, EP		NA
	Tris hydrochloric acid	USP, EP		NA
	Urea	NA		Non-product contact
Purification filters	Pellicon 3 Ultracel C Screen 5 kDa MWCO	NA	P3C005C01 P3C005C05 P3C005C10	0.11 m^2 0.57 m^2 1.14 m^2
	Millipore Durapore 0.22 µm	NA	NA	Sterile Grade 0.2 µm
	Millipore Express SHC, 0.5 µm/0.2 µm	NA	NA	Generic 0.2 µm
	Pall Fluorodyne EX EDF, 0.2 µm	NA	NA	Generic 0.2 µm
Miscellaneous components	1 L HDPE Bottles	NA		NA
	4 L HDPE Bottles	NA		NA

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3 IN-PROCESS HOLD TIMES

Process intermediate stability studies were performed to support pooling of chromatography cycles and holding of process intermediates.

Hold times begin when product is collected and end when the next step is loaded.

Unless otherwise noted, room temperature hold times are additive to refrigerated temperature hold times but not vice versa. For example, a product that was held for 2 days at room temperature prior to being stored 1 day refrigerated would have an additive hold time of 3 days at refrigerated temperature but only 2 days at room temperature.

Table 3-1 Process Intermediate Hold Times

Process Intermediate	Room Temperature Hold Time - Max (days, 15-25 °C)	Refrigerated Temperature Hold Time - Max (days, 2-8 °C)	Justification/Notes
Conditioned medium	1	3	In-process hold data
CEX chromatography Product	1.5	3	
AEXMM chromatography load	1.5	3	
AEXMM chromatography product	1.5	3	
HIC Load	1.5	3	
HIC product	3	5	
UFDF product DS	2	5	

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4 LOAD AND PRODUCT FILTRATION

In this section, load and product filter strategies are presented for each unit operation. Under each "**Unit Operation**", the order of filtration steps is described in reference to the unit operation. Each "**Filtration Step**" has its associated filter choices listed in the right hand "**Membrane Type/Choices**" column. Only one choice of membrane type should be made for each "**Filtration Step**".

Unit Operation	Filtration Step	Membrane Type/Choices	Grade/ Pore Size (µm)
CEX chromatography (Load 1 → Unit Op → Product)	Load 1	Millipore SHC	0.5/0.2
		Pall Fluorodyne EX EDF	0.2
	Product	Millipore SHC	0.5/0.2
		Pall Fluorodyne EX EDF	0.2
AEXMM Chromatography (Load → Unit Op → Product)	Load	Millipore SHC	0.5/0.2
		Pall Fluorodyne EX EDF	0.2
	Product	Millipore SHC	0.5/0.2
		Pall Fluorodyne EX EDF	0.2
HIC Chromatography (Load → Unit Op → Product)	Load	Millipore SHC	0.5/0.2
		Pall Fluorodyne EX EDF	0.2
	Product	Millipore SHC	0.5/0.2
		Pall Fluorodyne EX EDF	0.2
Concentration and Diafiltration (Unit Op → Product)	Product	Millipore Durapore	0.22

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5 CATION EXCHANGE CHROMATOGRAPHY

Step	Parameter (units)	Target	NOR		Justification/Notes
			Min	Max	
General	Media	GigaCap CM-650M			Prep per appropriate SOP
	Load filter in-line (Y/N)	Y	NA	NA	SHC
Pre-Equilibration *32.5 mM sodium citrate, 400 mM NaCl, pH 5.5*	Linear velocity (cm/hr)	130	NA	150	
	Volume (CV)	3.0	2.9	4.0	
	pH [a]	NA	5.4	5.6	Effluent, final
Equilibration 1 *32.5 mM sodium citrate, pH 5.5*	Linear velocity (cm/hr)	130	NA	150	
	Volume (CV)	3.0	2.9	4.0	
	pH [a]	NA	5.4	5.6	Effluent, final
Load *Conditioned Medium + WFI (in-line dilution)*	Linear velocity (cm/hr)	150	NA	NA	This is based on pump A (CM) operating at 13% and pump B (water) operating at 87% (at max pump speed)
	Buffer B % (WFI)	87/89	86	89	In-line dilution with water Cond ≥ 26.6 mS/cm, 89% water Cond < 26.6 mS/cm, 87% water
	Protein load (g/L resin) (by RP-HPLC assay)	15	10	20	CM pH should be between 5.2 to 5.4 before in-line dilution
	Conductivity (mS/cm) [a]	NA	NA	5.5	Effluent, final. Run and record.
Equilibration 2 *32.5 mM sodium citrate, pH 5.5*	Linear velocity (cm/hr)	130	NA	150	
	Volume (CV)	5.0	4.9	6.0	
Elution (LG) *32.5 mM sodium citrate, 400 mM NaCl, pH 5.5*	Linear velocity (cm/hr)	130	NA	150	
	Volume (CV)	6.4	NA	NA	
	Buffer B %, initial	10	NA	NA	
	Buffer B %, final	90	NA	NA	
	Start collection mAU	200	180	220	Volume before peak = ~ 1.7 CV

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Step	Parameter (units)	Target	NOR		Justification/Notes
			Min	Max	
	End collection mAU	200	180	220	Product volume = ~2-3 CV
Post elution hold	Linear velocity (cm/hr)	130	NA	150	
32.5 mM sodium citrate, 400 mM NaCl, pH 5.5	Buffer B %	100	NA	NA	
	Volume (CV)	2.0	1.9	3.0	
Strip 1	Linear velocity (cm/hr)	130	NA	150	
1 M NaCl	Volume (CV)	2.0	1.9	3.0	
Strip 2	Linear velocity (cm/hr)	130	NA	150	
100 mM Arg, 5 M Urea	Volume (CV)	2.0	1.9	3.0	
Strip 3	Linear velocity (cm/hr)	130	NA	150	
WFI	Volume (CV)	2.0	1.9	3.0	
Sanitization	Linear velocity (cm/hr)	130	NA	150	
0.5 N NaOH	Volume (CV)	2.0	1.9	3.0	Sanitization buffer
Sanitization Hold	Contact time (minutes)	15	NA	30	
Pre-Storage Flush	Linear velocity (cm/hr)	130	NA	150	
32.5 mM sodium citrate, 400 mM NaCl, pH 5.5	Volume (CV)	3.0	2.5	NA	
	pH [a]	NA	5.2	5.8	Effluent, final
Storage	Linear velocity (cm/hr)	130	NA	150	
20% (v/v) Ethanol	Volume (CV)	3.0	2.5	4.0	

[a] Output parameter

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6 AEXMM CHROMATOGRAPHY LOAD ADJUSTMENT

Step	Parameter (units)	Target	NOR		Justification/Notes
			Min	Max	
Adjustment *100 mM Tris, 3.2 M ammonium sulfate, pH 8.5*	Kg of adjustment buffer/kg of CEX elution	0.0383	NA	NA	The load adjustment is intended to bring the CEX elution to 100 mM ammonium sulfate concentration
Adjustment *1 M Tris*	pH	8.5	8.4	8.6	~0.09 (kg titrant /kg CEX Product)

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7 AEXMM CHROMATOGRAPHY

Step	Parameter (units)	Target	NOR		Justification/Notes
			Min	Max	
General	Media	Capto Adhere ImpRes			Prep per appropriate SOP
	Load filter in-line (Y/N)	Y	NA	NA	SHC
Equilibration 1 *100 mM tris, 10 mM ammonium sulfate, pH 8.5*	Linear velocity (cm/hr)	100	NA	150	
	Volume (CV)	5.0	4.9	6.0	
	pH [a]	NA	8.3	8.7	Effluent
	Conductivity (mS/cm) [a]	NA	NA	NA	Effluent, final. Run and record.
Load *Adjusted CEX elution + WFI (in-line dilution)*	Linear velocity (cm/hr)	100	NA	150	
	Buffer B % (WFI)	90	89	90	In-line dilution with water
	Protein load (g/ L resin) (by RP-HPLC assay)	9	6	12	
	Conductivity (mS/cm) [a]	NA	NA	6.0	Effluent, final. Run and record.
Equilibration 2 *100 mM tris, 10 mM ammonium sulfate, pH 8.5*	Linear velocity (cm/hr)	100	NA	150	
	Volume (CV)	5.0	4.9	6.0	
Elution (LG) *100 mM Tris, 10 mM ammonium sulfate, 250 mM magnesium chloride, pH 8.5*	Linear velocity (cm/hr)	100	NA	150	
	Volume (CV)	4	NA	NA	
	Buffer B %, initial	10	NA	NA	
	Buffer B %, final	90	NA	NA	
	Start collection mAU	100	85	120	Volume before peak = ~ 1.1 CV
	End collection mAU	650	NA	670	Product volume = ~1-2 CV
Post elution hold *100 mM Tris, 10 mM ammonium sulfate, 250 mM magnesium chloride, pH 8.5*	Linear velocity (cm/hr)	100	NA	150	
	Buffer B %	100	NA	NA	
	Volume (CV)	2.0	1.9	3.0	

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Step	Parameter (units)	Target	NOR		Justification/Notes
			Min	Max	
Strip 1	Linear velocity (cm/hr)	100	NA	150	
WFI	Volume (CV)	2.0	1.9	3.0	
Strip 2	Linear velocity (cm/hr)	100	NA	150	
100 mM acetic acid	Volume (CV)	2.0	1.9	3.0	
Strip 3	Linear velocity (cm/hr)	100	NA	150	
WFI	Volume (CV)	2.0	1.9	3.0	
Sanitization	Linear velocity (cm/hr)	100	NA	150	
1 N sodium hydroxide	Volume (CV)	3.0	2.9	4.0	Sanitization buffer
Sanitization Hold	Contact time (minutes)	15	NA	30	
Pre-Storage Flush *100 mM Tris, 10 mM ammonium sulfate, pH 8.5*	Linear velocity (cm/hr)	150	NA	160	
	Volume (CV)	3.0	2.9	4.0	
	pH [a]	NA	8.2	8.8	Effluent, final
Storage	Linear velocity (cm/hr)	150	NA	160	
20% Ethanol	Volume (CV)	3.0	2.5	4.0	

[a] Output parameter

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8 HIC LOAD ADJUSTMENT

Step	Parameter (units)	Target	NOR		Justification/Notes
			Min	Max	
Adjustment *100 mM Tris, 3.2 M ammonium sulfate, pH 8.5*	Kg of adjustment buffer/kg of AEXMM elution	1.98	NA	NA	The load adjustment is intended to bring the CAI elution to 2 M ammonium sulfate concentration

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9 HIC CHROMATOGRAPHY

Step	Parameter (units)	Target	NOR		Justification/Notes
			Min	Max	
General	Media	Capto Phenyl (high sub)			Prep per appropriate SOP
	Load filter in-line (Y/N)	Y	NA	NA	SHC
Equilibration 1 *100 mM Tris, 2 M ammonium sulfate, pH 8.5*	Linear velocity (cm/hr)	150	NA	170	
	Volume (CV)	5.0	4.9	6.0	
Load *AEXMM Chromatography Load Adjustment Product*	Linear velocity (cm/hr)	150	NA	170	
	Protein load (g / L resin)	9	6	12	
Equilibration 2 *100 mM Tris, 2 M ammonium sulfate, pH 8.5*	Linear velocity (cm/hr)	150	NA	170	
	Volume (CV)	3.0	2.9	4.0	
Wash *30%B (100 mM Tris, 1.4 M ammonium sulfate, pH 8.5)*	Linear velocity (cm/hr)	150	NA	170	
	Volume (CV)	2.0	NA	NA	
	Buffer B %	30	NA	NA	
Elution(LG) *100 mM Tris, pH 8.5*	Linear velocity (cm/hr)	150	NA	170	
	Volume (CV)	12.0	NA	NA	
	Buffer B %, initial	30	NA	NA	
	Buffer B %, final	90	NA	NA	
	Start collection mAU	50	45	55	Volume before peak = ~ 2.5 CV
	End collection mAU	30% PM	29% PM	31% PM	Product volume = ~5-8 CV
Post elution hold *100 mM Tris, pH 8.5*	Linear velocity (cm/hr)	150	NA	170	
	Buffer B %	100	NA	NA	
	Volume (CV)	2.0	1.9	3.0	
Strip 1	Linear velocity (cm/hr)	150	NA	170	

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Step	Parameter (units)	Target	NOR		Justification/Notes
			Min	Max	
WFI	Volume (CV)	2.0	1.9	3.0	
Strip 2	Linear velocity (cm/hr)	150	NA	170	
100 mM arginine, 5 M Urea	Volume (CV)	2.0	1.9	3.0	
Strip 3	Linear velocity (cm/hr)	150	NA	170	
WFI	Volume (CV)	2.0	1.9	3.0	
Sanitization	Linear velocity (cm/hr)	150	NA	170	
1 N sodium hydroxide	Volume (CV)	3.0	2.9	4.0	Sanitization buffer
Sanitization Hold	Contact time (minutes)	15	NA	30	
Pre-Storage Flush *100 mM Tris, pH 8.5*	Linear velocity (cm/hr)	150	NA	170	
	Volume (CV)	3.0	2.9	4.0	
	pH [a]	NA	8.2	8.8	Effluent, final
Storage	Linear velocity (cm/hr)	150	NA	170	
20% Ethanol	Volume (CV)	3.0	2.9	4.0	

[a] Output parameter

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10 CONCENTRATION AND DIAFILTRATION

Step	Parameter (units)	Target	NOR		Justification/Notes
			Min	Max	
General	Media	Pellicon 3 Ultracel C Screen 5kDa MWCO			Prep per approriate SOP
	Feed pressure (psig)	NA	NA	NA	
Equilibration *4 mM Sodium/Potassium Phosphate, 155 mM Sodium Chloride, pH 7.0*	pH [a]	7.0	6.8	7.2	
Concentration 1 Control Method: TMP and Feed Flow Control	Membrane protein load (g/m^2)	200	NA	300	(Mass of protein)/(area of ultrafiltration membrane)
	TMP (psig)	30	NA	35	
	Feed flow (L/min/m^2)	5.0	4.0	6.0	
	Conc. 1, final (g/L)	24	20	30	May be directly measured or calculated from permeate or retentate volume
Diafiltration *4 mM Sodium/Potassium Phosphate, 155 mM Sodium Chloride, pH 7.0* Control Method: TMP and Feed Flow Control	TMP (psig)	30	NA	35	
	Feed flow (L/min/m^2)	5.0	4.0	6.0	
	Diafiltration volumes	8.0	NA	10.0	(kg buffer)/(kg product)
	Permeate pH, final [a]	7.0	6.8	7.2	
	Retentate pH, final [a]	7.0	6.8	7.2	
Concentration 2 Control Method: TMP and Feed Flow Control	TMP (psig)	30	NA	35	
	Feed flow (L/min/m^2)	5.0	4.0	6.0	
	Conc. 2, final (g/L)	75	70	85	May be directly measured or calculated from permeate or retentate volume

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Step	Parameter (units)	Target	NOR		Justification/Notes
			Min	Max	
Recirculation	Feed pressure (psig)	20	NA	40	
	Feed flow (L/min/m^2)	5.0	4.0	6.0	
	Recirculation time (min)	5	NA	30	
Recovery Flush	Feed pressure (psi)	20	NA	40	
4 mM Sodium/Potassium Phosphate, 155 mM Sodium Chloride, pH 7.0	Feed flow (L/min/m^2)	5.0	4.0	6.0	
Recovery Method: Buffer flush of system hold up volume	Recovery conc. (g/L)	65	60	70	
Dilution to DS	DS conc. (g/L)	55	50	60	
4 mM Sodium/Potassium Phosphate, 155 mM Sodium Chloride, pH 7.0	pH [a]	7.0	6.8	7.2	

[a] Output parameter

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11 BULK FILTRATION

Step	Parameter (units)	Target	NOR		Justification/Notes
			Min	Max	
0.2µm Bulk Filtration *Millipore Durapore*	Capacity (L/m^2)	NA	NA	NA	
	Drug substance container	1L or 4L PETG bottles			
	Fill volume percentage (%)	70	65	75	If multiple containers are required, target equivalent volume in each.
	Drug substance storage temperature (°C)	- 80	NA	- 65	
Integrity Test *0.2 µm Filter*	Integrity Test Result (PASS/FAIL)	PASS	PASS	PASS	Perform per appropriate SOP. Integrity test is an in-process control that must pass for the lot to be released

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12 BUFFERS AND SOLUTIONS

Buffer Name	Recipe Table	Unit Operation	Steps
32.5 mM Sodium Citrate, pH 5.5	Table 12-1	CEX chromatography	Equilibration 1
			Equilibration 2
			Elution gradient
32.5 mM Sodium Citrate, 400 mM Sodium Chloride, pH 5.5	Table 12-2	CEX Chromatography	Pre-Eqilibration
			Elution gradient
			Post elution hold
			Pre-Storage flush
1 M sodium chloride	Table12-3	CEX Chromatography	Strip 1
100 mM Arginine, 5M Urea	Table 12-4	CEX Chromatography	Strip 2
		HIC Chromatography	Strip 2
100 mM Tris, 10 mM Ammonium Sulfate, pH 8.5	Table 12-5	AEXMM Chromatography	Equilibration 1
			Equilibration 2
			Elution gradient
100 mM Tris, 10 mM Ammonium Sulfate, 250 mM Magnesium Chloride, pH 8.5	Table 12-6	AEXMM Chromatography	Elution gradient
			Post elution hold
			Pre-Storage flush
100 mM Tris, 3.2 M Ammonium Sulfate, pH 8.5	Table 12-7	AEXMM Chromtography	AEXMM Load adjustment
		HIC Chromatography	HIC Load adjustment
100 mM Tris, 2 M Ammonium Sulfate, pH 8.5	Table 12-8	HIC Chromatography	Equilibration 1
			Equilibration 2
			Elution gradient
100 mM Tris, pH 8.5	Table 12-9	HIC Chromatography	Elution gradient
			Post elution hold
			Pre-Storage flush
100 mM acetic acid	Table 12-10	AEX Chromatography	Strip 2

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Buffer Name	Recipe Table	Unit Operation	Steps
WFI	-	CEX Chromatography	Load, in-line dilution
			Strip 3
		AEXMM Chromatography	Load, in-line dilution
			Strip 1
			Strip 3
		HIC Chromatography	Strip 1
			Strip 3
0.5 N sodium hydroxide	Lonza standard recipe	CEX Chromatography	Sanitization
20% Ethanol	Lonza standard recipe	CEX Chromatography	Storage
		AEXMM Chromatography	Storage
		HIC Chromatography	Storage
1 M tris base	Lonza standard recipe	AEXMM chromatography	Load pH adjustment
1 N sodium hydroxide	Lonza standard recipe	AEXMM Chromatography	Sanitization
		HIC Chromatography	Sanitization
4 mM Sodium/Potassium Phosphate, 155 mM Sodium Chloride, pH 7.0	Table 12-11	UFDF	Equilibration
			Diafiltration
			Recirculation
			Dilution to DS

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Table 12-1

32.5 mM Sodium Citrate, pH 5.5

Density Corrected	Yes		
Specification	**Target**	**Operating Range**	
pH	5.5	5.4 – 5.6	
Conductivity (mS/cm at 25°C)	5.5	5.23 – 5.78	
Osmolality (mOsm/kg)	NA	NA	
Density (g/mL)	1.000	NA	
Chemical	**FW**	**Amount by Volume**	**Amount by Weight**
Citric Acid, Monohydrate	210.14 g/mol	NA	1.67 g/kg
Sodium Citrate, Dihydrate	294.10 g/mol	NA	7.23 g/kg
WFI	NA	NA	991.1 g/kg
Total		**1000 g/kg**	

Table 12-2

32.5 mM Sodium Citrate, 400 mM Sodium Chloride, pH 5.5

Density Corrected	Yes		
Specification	**Target**	**Operating Range**	
pH	5.5	5.4 – 5.6	
Conductivity (mS/cm at 25°C)	42	39.9 – 44.1	
Osmolality (mOsm/kg)	NA	NA	
Density (g/mL)	1.019	NA	
Chemical	**FW**	**Amount by Volume**	**Amount by Weight**
Citric Acid, Monohydrate	210.14 g/mol	NA	0.87 g/kg
Sodium Citrate, Dihydrate	294.10 g/mol	NA	8.16 g/kg
Sodium Chloride	58.44 g/mol	NA	22.94 g/kg
WFI	NA	NA	968.03 g/kg
Total		**1000 g/kg**	

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Table 12-3

1 M sodium chloride

Density Corrected	Yes		
Specification	**Target**	**Operating Range**	
pH	NA	NA	
Conductivity (mS/cm at 25°C)	82	78-86	
Osmolality (mOsm/kg)	NA	NA	
Density (g/mL)	1.03	NA	
Chemical	**FW**	**Amount by Volume**	**Amount by Weight**
Sodium chloride	58.44	NA	56.74 g/kg
WFI	NA	NA	943.26 g/kg
Total		**1000 g/kg**	

Table 12-4

100 mM Arginine, 5M Urea

Density Corrected	Yes		
Specification	**Target**	**Operating Range**	
pH	NA	11.2 (expected)	
Conductivity (mS/cm at 25°C)	NA	0.2 (expected)	
Osmolality (mOsm/kg)	NA	NA	
Density (g/mL)	1.078	NA	
Chemical	**FW**	**Amount by Volume**	**Amount by Weight**
L-Arginine	174.20 g/mol	NA	16.16 g/kg
Urea	60.06 g/mol	NA	278.57 g/kg
WFI	NA	NA	705.27 g/kg
Total		**1000 g/kg**	

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Table 12-5

100 mM Tris, 10 mM Ammonium Sulfate, pH 8.5

Density Corrected	Yes		
Specification	**Target**	**Operating Range**	
pH	8.5	8.3 – 8.7	
Conductivity (mS/cm at 25°C)	4.9	4.66 – 5.15	
Osmolality (mOsm/kg)	NA	NA	
Density (g/mL)	1.001	NA	
Chemical	**FW**	**Amount by Volume**	**Amount by Weight**
Ammonium Sulfate	132.14 g/mol	NA	1.32 g/kg
Tris base	121.14 g/mol	NA	8.33 g/kg
Tris Hydrochloride	157.60 g/mol	NA	4.90 g/kg
WFI	NA	NA	985.45 g/kg
Total		1000 g/kg	

Table 12-6

100 mM Tris, 10 mM Ammonium Sulfate, 250 mM Magnesium Chloride, pH 8.5

Density Corrected	Yes		
Specification	**Target**	**Operating Range**	
pH	8.5	8.3 – 8.7	
Conductivity (mS/cm at 25°C)	42	39.9– 44.1	
Osmolality (mOsm/kg)	NA	NA	
Density (g/mL)	1.021	NA	
Chemical	**FW**	**Amount by Volume**	**Amount by Weight**
Ammonium Sulfate	132.14 g/mol	NA	1.29 g/kg
Magnesium Chloride, Hexahydrate	203.30 g/mol	NA	49.78 g/kg
Tris base	121.14 g/mol	NA	8.43 g/kg
Tris Hydrochloride	157.60 g/mol	NA	4.47 g/kg
WFI	NA	NA	936.03 g/kg
Total		1000 g/kg	

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Table 12-7

100 mM Tris, 3.2 M Ammonium Sulfate, pH 8.5

Density Corrected	Yes		
Specification	**Target**	**Operating Range**	
pH	8.5	8.3 – 8.7	
Conductivity (mS/cm at 25°C)	260	247 – 273	
Osmolality (mOsm/kg)	NA	NA	
Density (g/mL)	1.201	NA	
Chemical	**FW**	**Amount by Volume**	**Amount by Weight**
Tris base	121.14 g/mol	NA	10.08 g/kg
Ammonium Sulfate	132.14 g/mol	NA	352.08 g/kg
WFI	NA	NA	637.84 g/kg
Total		1000 g/kg	

Table 12-8

100 mM Tris, 2 M Ammonium Sulfate, pH 8.5

Density Corrected	Yes		
Specification	**Target**	**Operating Range**	
pH	8.5	8.3 – 8.7	
Conductivity (mS/cm at 25°C)	212	202-222	
Osmolality (mOsm/kg)	NA	NA	
Density (g/mL)	1.135	NA	
Chemical	**FW**	**Amount by Volume**	**Amount by Weight**
Ammonium Sulfate	132.14 g/mol	NA	232.86 g/kg
Tris base	121.40 g/mol	NA	10.67 g/kg
10 N Sodium Hydroxide	40.00 g/mol	NA	1.60 g/kg
WFI	NA	NA	754.87 g/kg
Total		1000 g/kg	

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Table 12-9

100 mM Tris, pH 8.5

Density Corrected	Yes		
Specification	**Target**	**Operating Range**	
pH	8.5	8.3 – 8.7	
Conductivity (mS/cm at 25°C)	2.65	2.5 - 2.8	
Osmolality (mOsm/kg)	NA	NA	
Density (g/mL)	1.000	NA	
Chemical	**FW**	**Amount by Volume**	**Amount by Weight**
Tris base	121.14 g/mol	NA	8.46 g/kg
Tris Hydrochloride	157.60 g/mol	NA	4.75 g/kg
WFI	NA	NA	986.79 g/kg
Total		1000 g/kg	

Table 12-10

100 mM acetic acid

Density Corrected	Yes		
Specification	**Target**	**Operating Range**	
pH	Record	3.1-0	
Conductivity (mS/cm at 25°C)	Record	NA	
Osmolality (mOsm/kg)	NA	NA	
Density (g/mL)	1.000	NA	
Chemical	**FW**	**Amount by Volume**	**Amount by Weight**
Acetic acid, glacial	60.06 g/mol	NA	6.01 g/kg
WFI	NA	NA	993.99 g/kg
Total		1000 g/kg	

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Table 12-11

4 mM Sodium/Potassium Phosphate, 155 mM Sodium Chloride, pH 7.0

Density Corrected	Yes		
Specification	**Target**	**Operating Range**	
pH	7.0	6.9 – 7.1	
Conductivity (mS/cm at 25°C)	16.8	15.96 – 17.64	
Osmolality (mOsm/kg)	NA	NA	
Density (g/mL)	1.004	NA	
Chemical	**FW**	**Amount by Volume**	**Amount by Weight**
Potassium Phosphate Monobasic (KH_2PO_4)	136.09	NA	0.20 g/kg
Sodium Phosphate Dibasic Heptahydrate ($Na_2HPO_4 \cdot 7H_2O$)	268.07	NA	0.68 g/kg
Sodium Chloride	58.44 g/mol		9.06 g/kg
WFI	NA	NA	990.06 g/kg
Total		**1000 g/kg**	

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13 REFERENCE CHROMATOGRAMS

Sample chromatograms representing the three chromatography steps used in the purification process are presented below.

Table 13-1	**CEX Chromatogram**
Resin:	GigaCap CM-650M
Column Dimensions:	0.66 x 20 cm

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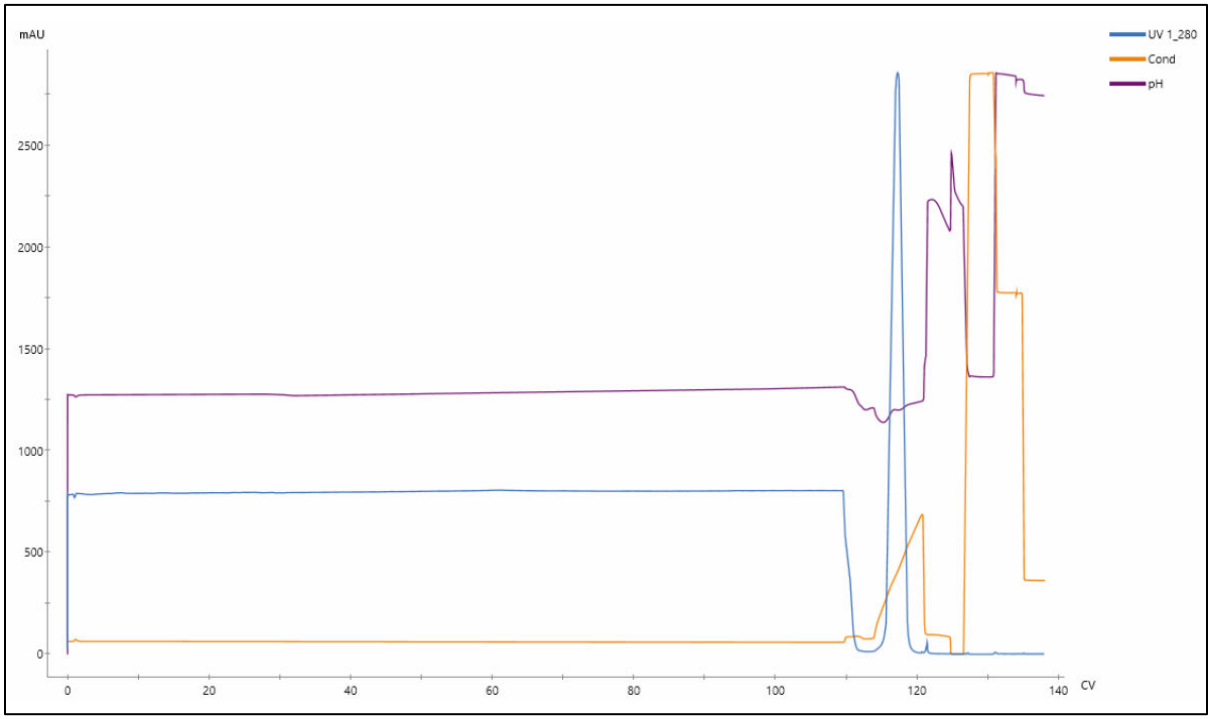

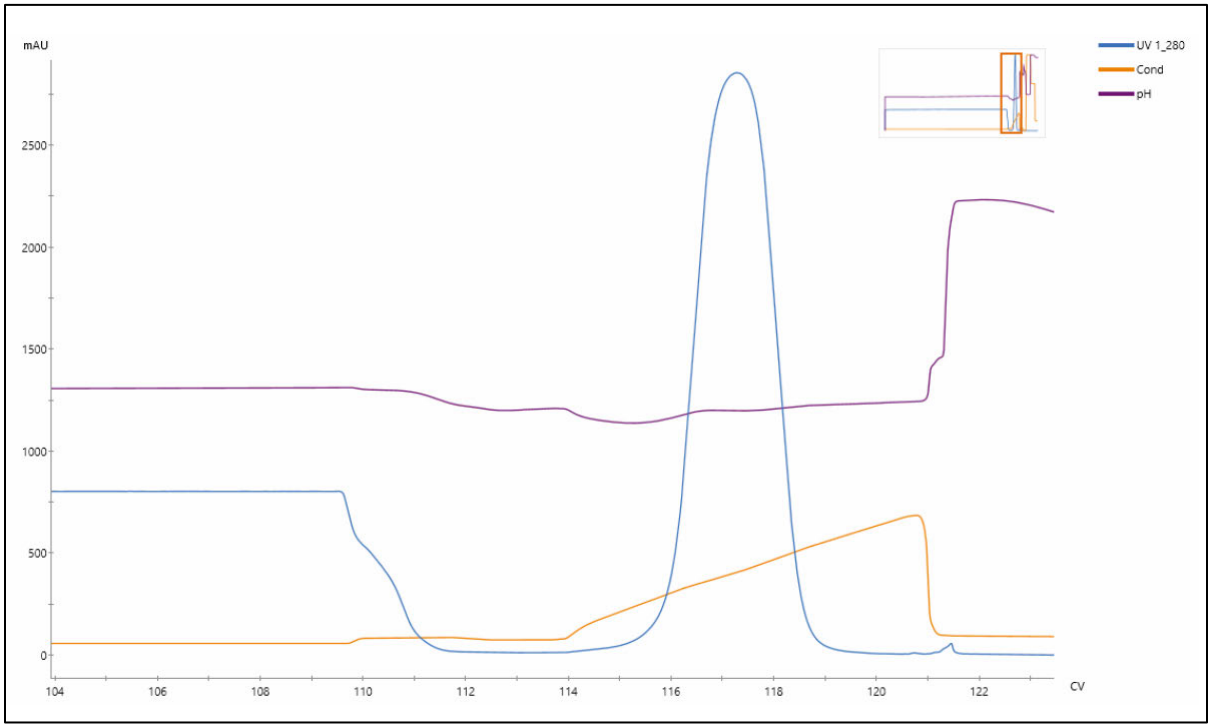

Figure 13-1 GigaCap CM-650M Chromatogram

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Table 13-2	**AEXMM Chromatogram**
Membrane:	Capto Adhere ImpRes
Column Dimensions:	0.66 x 20 cm

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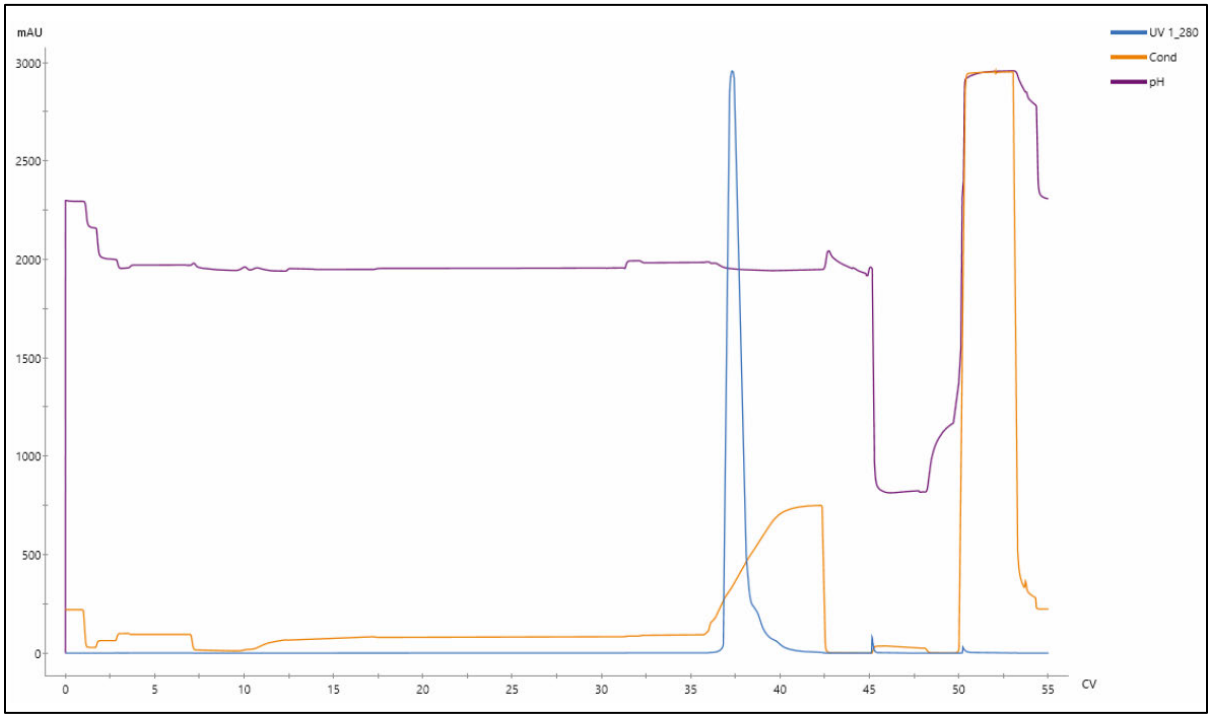

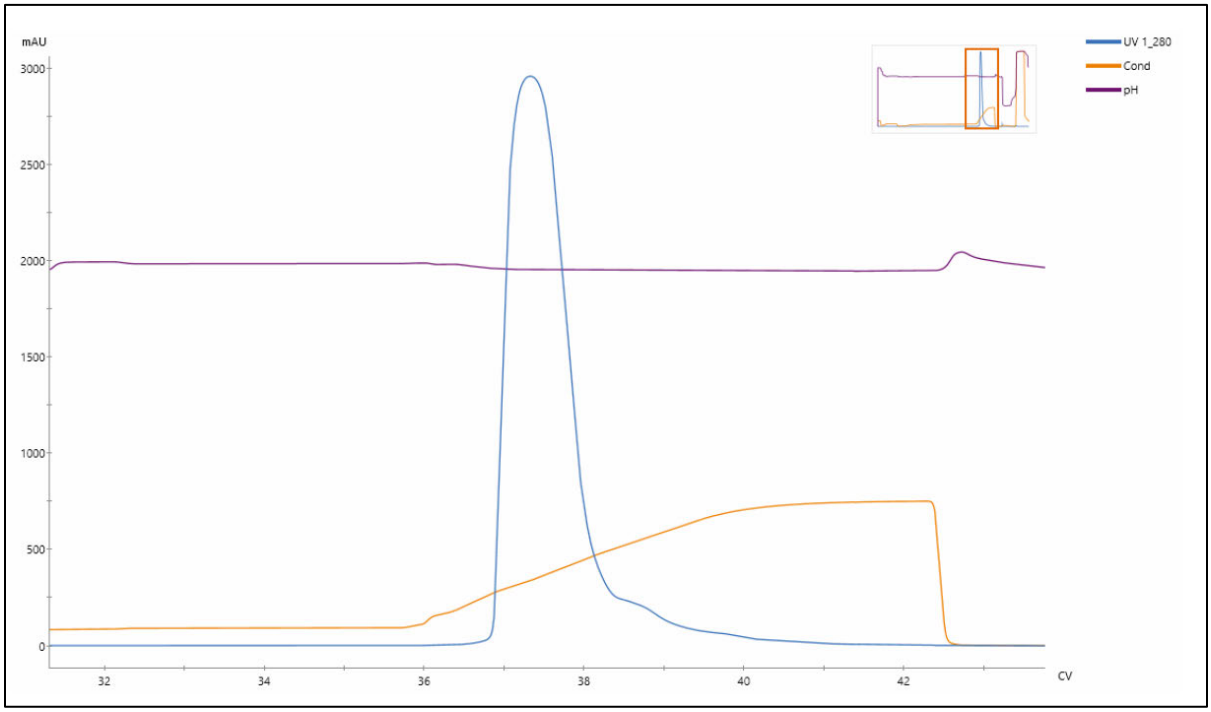

Figure 13-2 AEXMM Chromatogram

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Table 13-3	**HIC Chromatogram**
Resin:	Capto Phenyl (high sub)
Column Dimensions:	0.66 x 20.5 cm

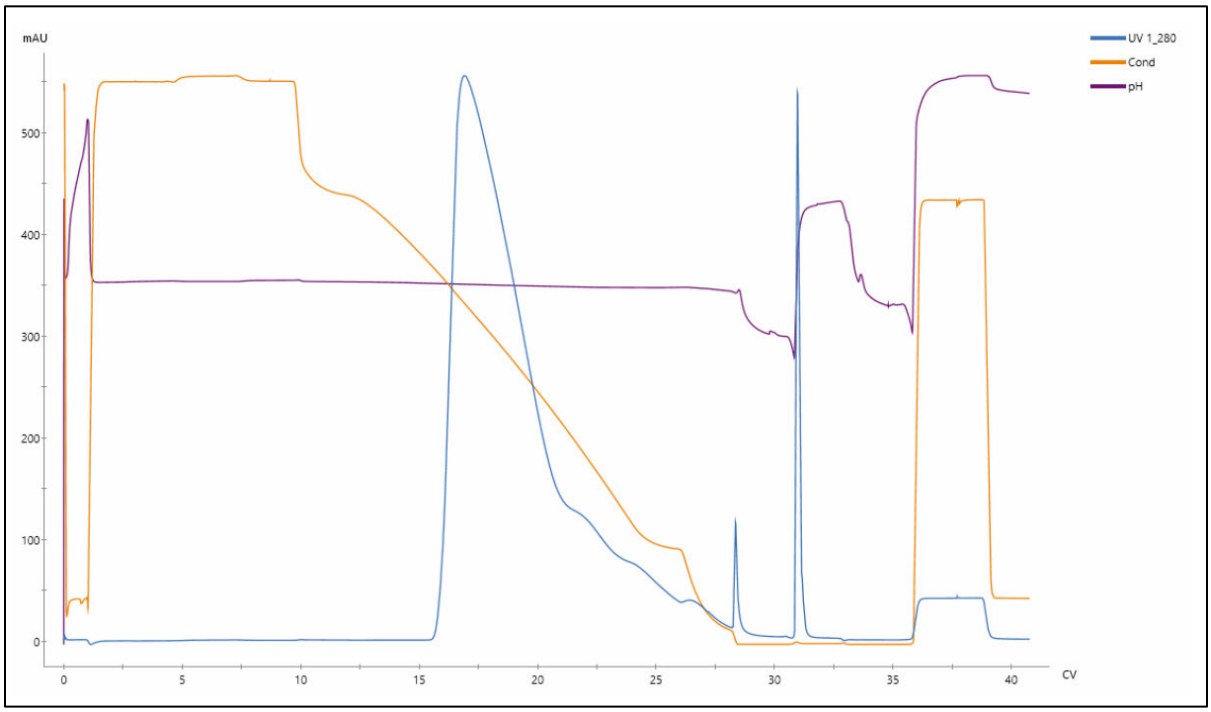

Figure 13-3 HIC Chromatogram

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14 EDITION HISTORY

Edition	Reason
001	New document

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